

Section 1: Executive Summary

1.1: The Challenge

In Equity Capital Markets (ECM), identifying actionable market mispricings requires analyzing hundreds of companies across multiple metrics. Currently, junior analysts spend several hours manually compiling Trailing P/E, Forward P/E, and EPS estimates from sources, calculating sector aggregates, and formatting data into presentable reports. This manual workflow is inefficient, prone to costly human error, and significantly limits the time available for high-level strategic client advisory.

1.2: The Solution

To resolve this bottleneck, our group engineered a fully automated Python-based data pipeline housed within a structured Jupyter Notebook. With a single click, this robust tool executes a live web-scraping script, cleanses the unstructured data, performs complex statistical aggregations, and instantly outputs a multi-sheet, conditionally formatted Excel dashboard ready for immediate executive review.

1.3: Methodology

Our pipeline programmatically extracts live financial fundamentals for all S&P 500 constituents using the Finviz screener. To ensure a highly accurate valuation baseline, the tool simultaneously scrapes over one hundred fifty years of historical S&P 500 data from Multpl.com. Rather than relying on simple arithmetic averages, the script dynamically calculates the historical median Trailing P/E. This effectively filters out extreme recessionary outliers, generating robust, mathematically sound "Above Average" and "Below Average" valuation bands.

1.4: Business Value

This automated solution transforms hours of tedious data entry into mere seconds of processing time. By instantly aggregating market-cap-weighted sector summaries and pinpointing overvalued or undervalued equities via a dedicated "Extremes Dashboard," ECM Directors can identify market trends, proactively source new investment opportunities, and provide superior, data-backed guidance to institutional clients.